PondGate user manual (v1.3.2)

1. What PondGate does and does not do

PondGate computes how water level, storage and outflow change over time in **one detention pond** with a **known inflow hydrograph**. It keeps a traceable record of every run.

- Method: the level-pool (storage-continuity) equation dS/dt = Qin − Qout(H), solved with a fixed-step fourth-order Runge–Kutta scheme (RK4, Δt = 2 s).
- It can: route one pond, report peak reduction and detention volume, run constrained screening and continuous optimization within declared limits, download records, and compare numerically with HEC-HMS or SWMM output.
- It cannot: compute rainfall–runoff, channel routing, long continuous simulation or multi-pond systems.
- Results are not a final engineering design and not a field calibration. Agreement with other models is called numerical consistency only. A qualified engineer remains responsible.
- When no feasible design exists, the software **blocks** the result and shows the reason and a correction path. It never shows "pass".

2. Screen layout

- **Left menu**: "Start here" is this help; "Setup and routing" holds Modules 1–4; "Constrained screening" is Module 5; "Results, verification and records" is Module 6; "Teaching and internal tests" holds a practice case and a short quiz.
- **Top right**: "English / 中文" switches the interface language. Input values do not change.
- **Bottom left "? Help"** opens the AI assistant and the next-step hint at any time.
- A **?** next to a field label explains that term.
- Every button press shows a short message at the bottom right.

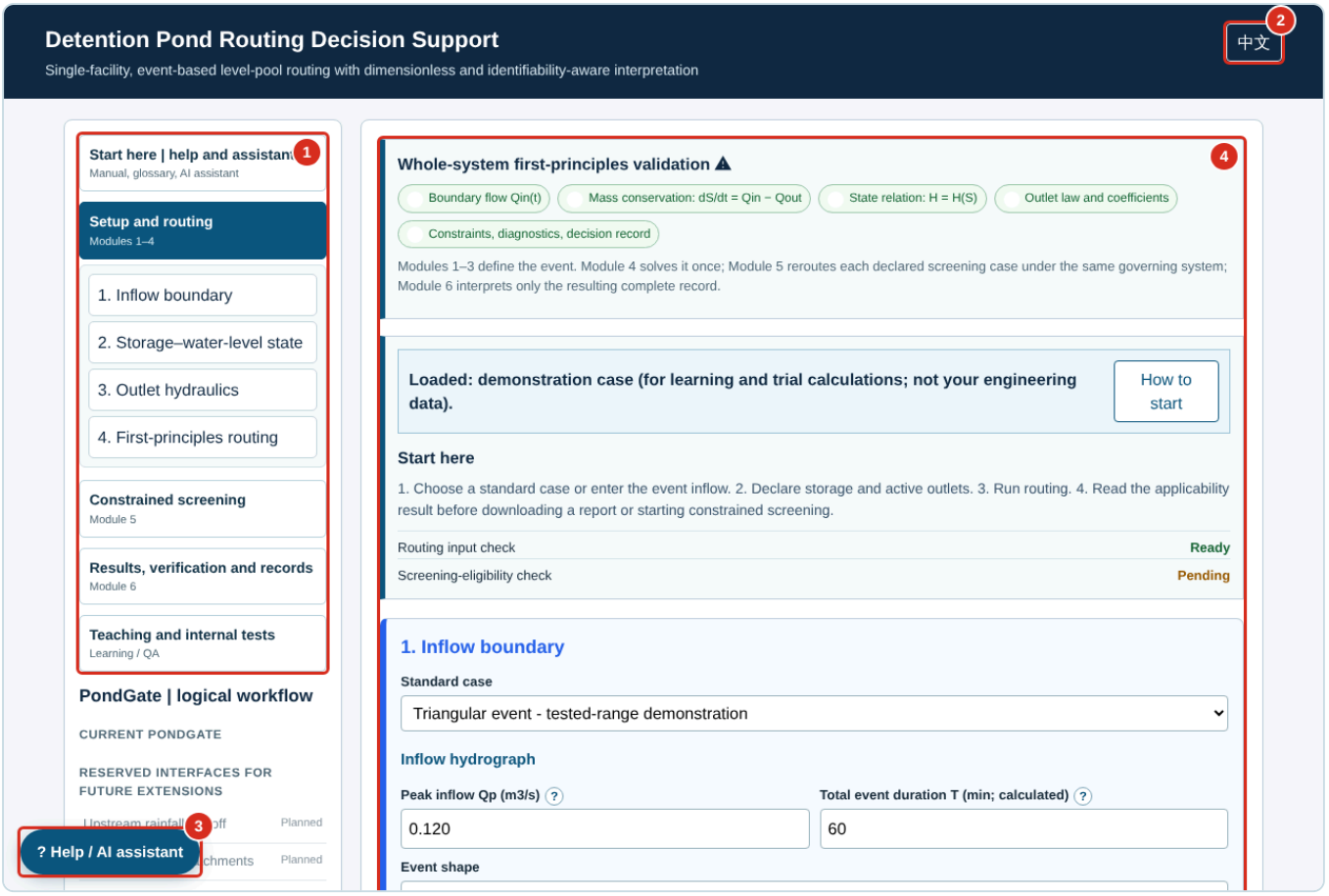


Figure 1. Screen layout

1. Left menu: start here, Modules 1–6, teaching
2. Language switch
3. Help / AI assistant button
4. Work area

3. Five-minute start with the demonstration case

1. The page opens with the "triangular event" demonstration case. It is not your engineering data.
2. Open "Setup and routing", scroll to the bottom and press **"Run routing and evaluation"**.
3. The page switches to "Results, verification and records". First check that the mass-balance residual is close to 0 %. Then read peak inflow, peak outflow and maximum water level.
4. To practise constrained screening, open "Teaching and internal tests", press "Load teaching example", then open "Constrained screening" and press "Run constrained design screening". The free-discharge ticks in the teaching case are practice assumptions only.
5. Press "Download reproducible decision-support report" to keep the record.

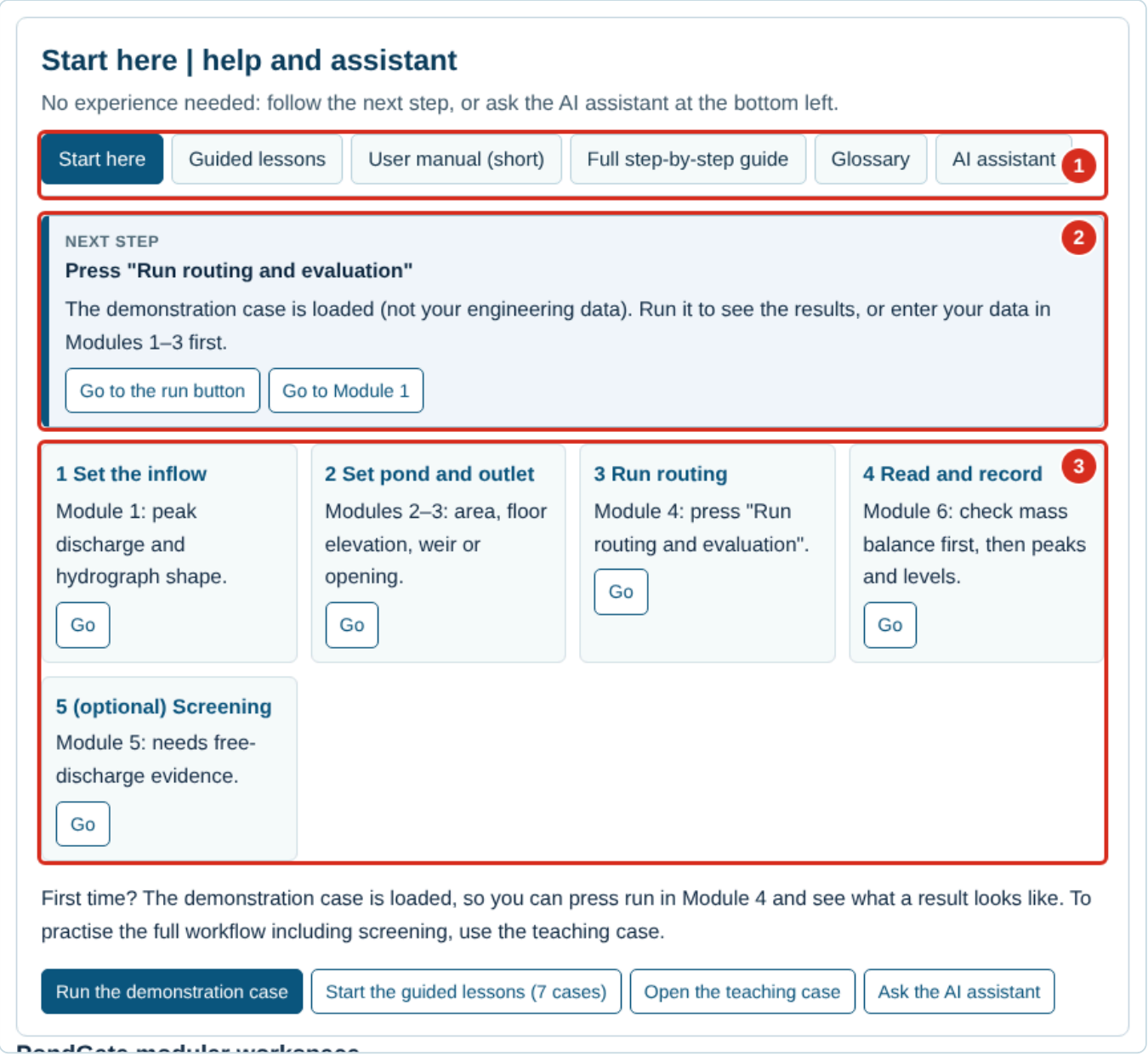


Figure 2. The "Start here" page

1. Tabs: start here, guided lessons, manual, glossary, AI assistant
2. Live next-step hint
3. Five steps with "Go" buttons

4. Guided lessons: learn the whole software in seven cases

"Start here → Guided lessons" holds seven lessons from simple to advanced. A lesson panel appears at the top right; each step is ticked automatically when done, then press "Next".

1. First routing run (demonstration case): run, read mass balance, download the report.

2. Your own event and pond: change Qp, Tp, Tr, area and weir width, and compare.
3. Long-duration event and peak timing: read FPI95 and use volume-time measures for a broad peak.
4. Find and fix a wrong input: the grey button and the red outline.
5. Constrained screening: complete free-discharge evidence and limits, get a feasible result, download the report.
6. No feasible solution: read the dominant limit and the right response.
7. Research-inspection mode: which outlet settings lock screening.

"Take me there" jumps to the field; "Show me" does the step for you. A lesson loads practice inputs and can restore yours at the end. Free-discharge ticks and records in the lessons are teaching assumptions, not for real projects.

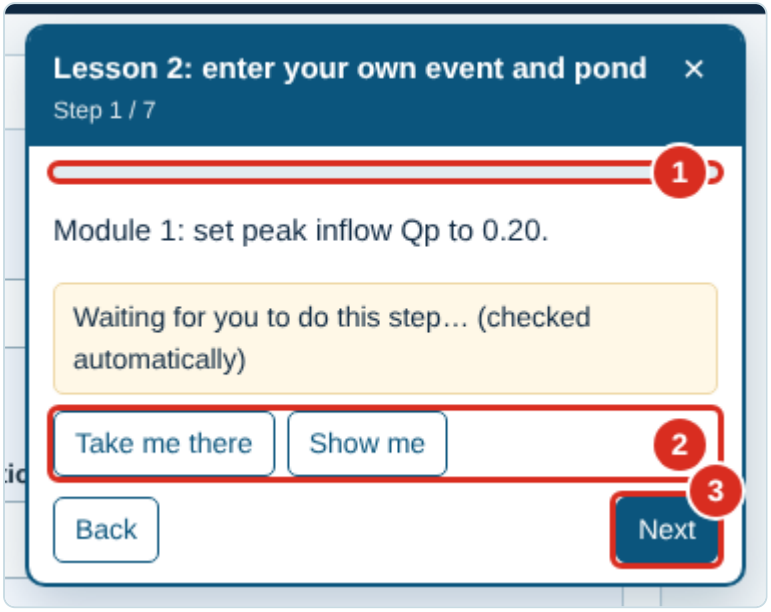


Figure 3. Guided-lesson panel

1. Progress bar and current step
2. "Take me there" and "Show me"
3. Press "Next" when the step is done

Teaching support | four-step self-learning

0/3 correct | Download practice case | Load teaching example (save current case) | Restore previous case

Define physical conditions, run the route, interpret the record, and then check screening eligibility. Teaching prompts never change the RK4 governing equation or convert the 176-case regression into universal design validation.

Teaching-case outlet relation Qout(H)

One sharp-crested weir: $Qout(H) = Cw \times b \times \max(H - P, 0)^{3/2}$

Cw = 1.70, b = 0.20 m, P = 0.00 m; at H = 0.50 m, Qout = 0.1202 m³/s. This relation applies only under the teaching free-discharge assumption.

1 Define event inflow

Choose a triangular or long-duration event. Duration and shape, not peak flow alone, affect detention storage.

Go to step

2 Declare storage and outlets

Use one vertical datum for S(H), the initial level, and every active outlet elevation.

Go to step

3 Run and interpret

Read mass balance, maximum level, peak outflow, and broad-crest timing interpretation first.

Go to step

4 Check screening eligibility

Constrained screening requires confirmed free discharge and a single passive weir or rectangular opening. It is not universal final-design validation.

Go to step

PondGate feature guide

Inflow boundary

Define the event peak, rise, plateau and recession; peak flow alone does not represent detention demand.

Storage–level state

Use constant area or tabulated S(H) with one consistent datum.

Outlet hydraulics

Declare passive outlets and coefficients; users must document tailwater, backwater, submergence and operation.

First-principles routing

Solve dS/dt = Qin − Qout with RK4 and inspect mass balance.

Constrained screening

Available only for one passive weir or one rectangular opening with complete free-discharge evidence; it is not final design.

Evaluation and support

Read attenuation, storage, timing and traceability records as the engineering decision basis.

The practice case is a teaching reconstruction of prior experimental settings; it excludes the 176-case raw regression data.

Quick self-check (about 30 seconds each)

1. Which term directly controls the time change of pond storage?

- ○ Inflow peak alone
- ○ Qin − Qout
- ○ Plan area alone

2. For the same inflow peak, what can still change detention demand?

- ○ Hydrograph duration and shape
- ○ Chart colour
- ○ Report filename

3. When is constrained design screening unavailable?

- ○ Confirmed free-discharge rectangular opening
- ○ Research-mode riser, pipe, or composite outlet
- ○ Validated passive weir

Figure 4. Teaching case on the teaching page

1. Load the teaching example that can run screening
2. Restore the previous case

5. Module 1: inflow boundary

- **Standard case**: triangular event, long-duration event, or user-defined event.
- **Peak inflow Qp** (m^3/s) must be greater than 0.
- **Inflow shape**: a triangle uses time to peak Tp and recession time Tr (or Tp and the ratio Tp/Tr); a long-duration plateau uses rising, plateau and recession times. The total event duration T is calculated and cannot be typed.
- **Imported time–discharge pairs**: one pair per line, "time (min), discharge (m^3/s)". The first time must be 0, times must increase, discharge must not be negative, and at least two lines are needed. Example: `0, 0`, `25, 0.10`, `50, 0.10`, `75, 0`.
- Peak discharge alone is not enough. Duration and shape also change the detention volume needed.

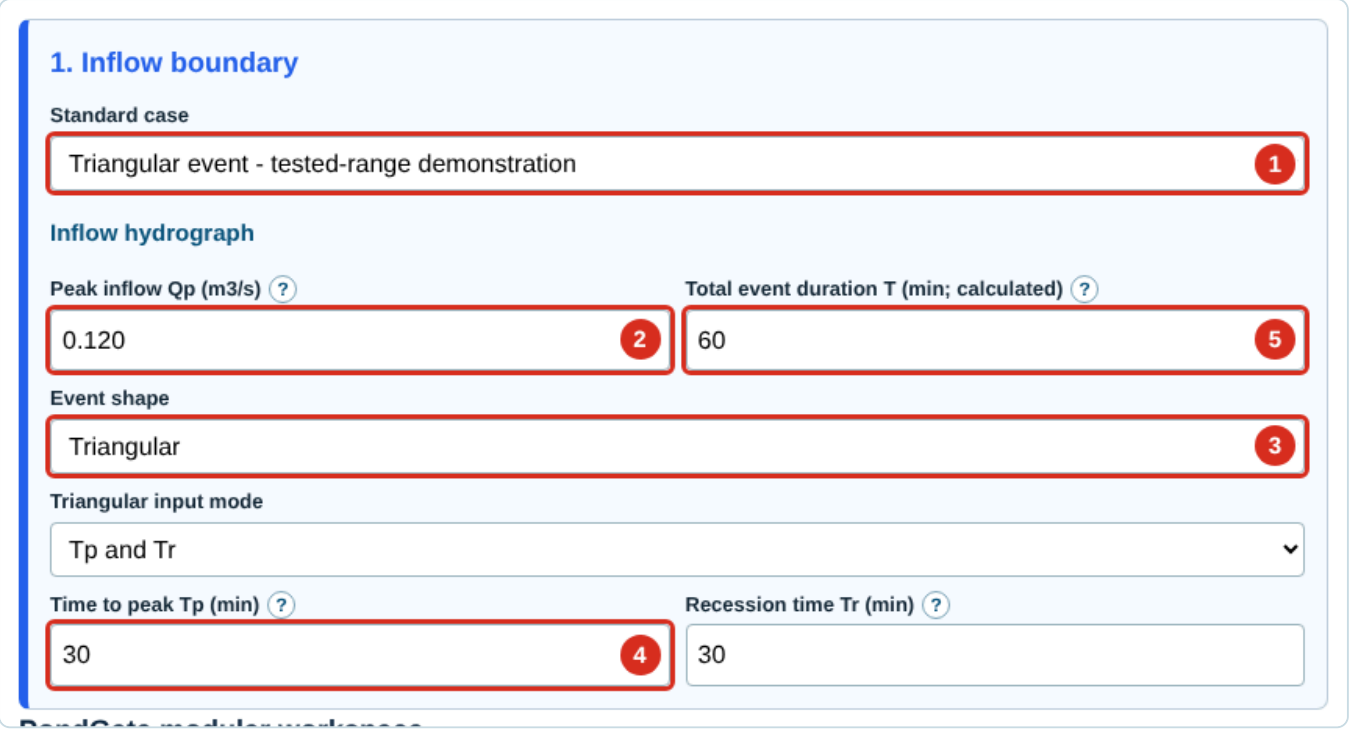


Figure 5. Module 1, inflow boundary

1. Standard case
2. Peak inflow Qp
3. Inflow shape
4. Time to peak Tp (and recession time Tr)
5. Total duration T (calculated)

6. Module 2: storage–water-level state

- **Constant plan area**: S(H) = A × (H − Hfloor). Only the plan area A (m^2, greater than 0) is needed.
- **Stage–storage table**: import a .csv or .txt file with two columns: water level (m), then storage (m^3). Routing does not extrapolate beyond the table.
- **Floor elevation Hfloor**, **initial level H0** and every outlet elevation must use **one vertical datum**. For a relative datum set Hfloor = 0. H0 must not be below Hfloor.

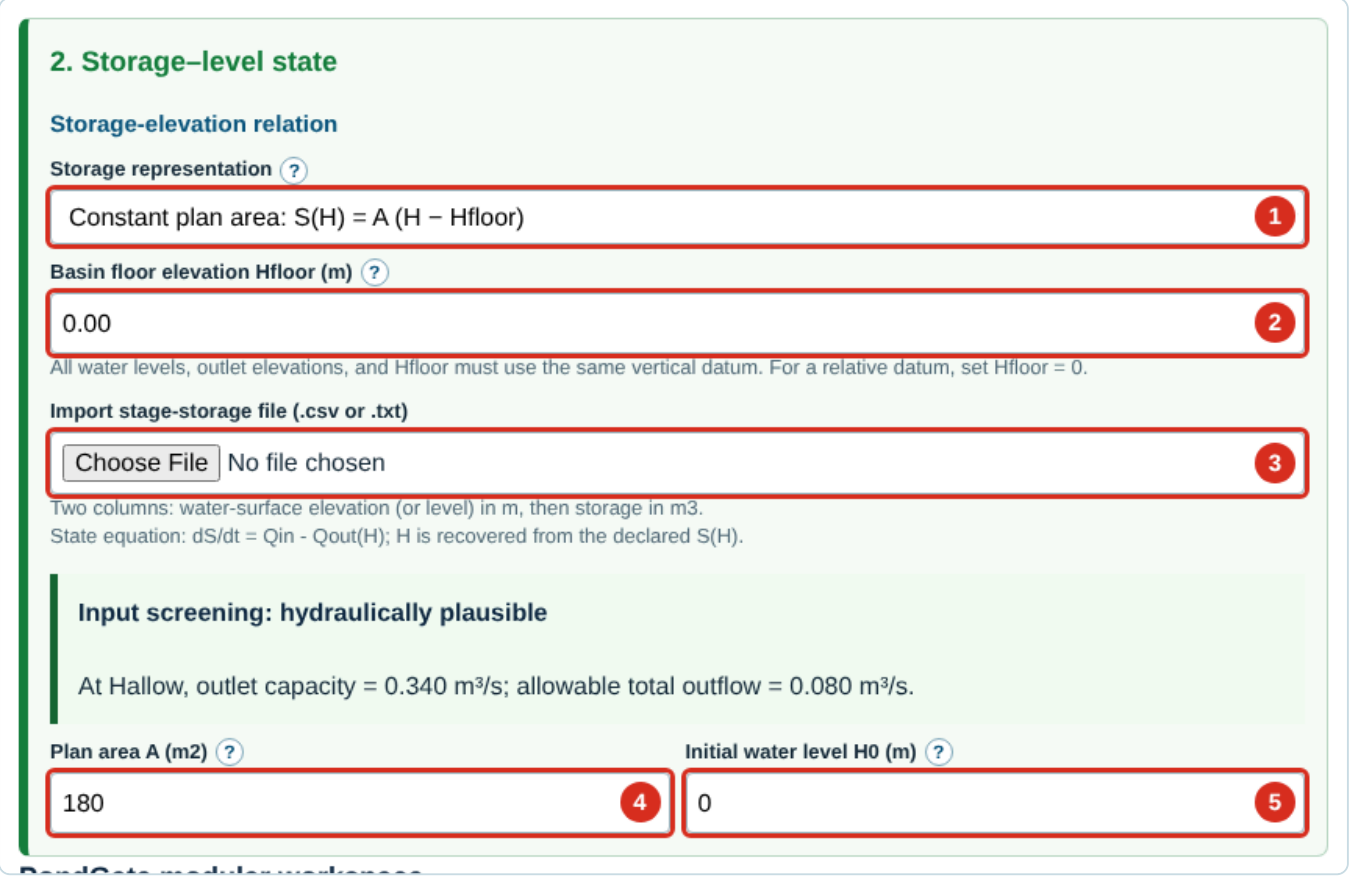


Figure 6. Module 2, storage–water-level state

1. Storage representation
2. Floor elevation Hfloor
3. Import a stage–storage table
4. Plan area A
5. Initial level H0

7. Module 3: outlet hydraulics and free-discharge evidence

- **Sharp-crested weir**: $Q_{out} = C_w \times b \times (H - P)^{1.5}$. Enter weir coefficient Cw, width b and crest elevation P.

- **Rectangular opening**: Qout = Cd × Ao × √(2g (H − invert)). Enter Cd, opening area Ao and invert elevation.
- **Riser, circular pipe, level-dependent coefficient table and weir + opening combinations** are "research inspection": routing works, but constrained screening is locked.
- Coefficients can use presets (for example Cw = 1.70, Cd = 0.62) or a field-calibrated value. Presets suit preliminary screening only.
- **Free-discharge evidence**: before screening, tick "Free-discharge boundary confirmed" and the four checks (tailwater, backwater or pipe friction, submergence, operational control), and fill in the two records (free-discharge evidence and screening-limit source). **Tick them only when you hold drawings, a field inspection or a calculation that supports them.** The software does not verify these conditions for you.

3. Outlet hydraulics

Outlet relation

- [ ] Vertical riser shaft

Outlet hydraulic boundary

- [ ] Free-discharge boundary confirmed (?) 4

Required evidence for constrained screening

- [ ] Tailwater remains below every controlling outlet.
- [ ] No downstream backwater or pipe-friction control modifies Qout(H).
- [ ] No outlet is submerged over the routed water-level range.
- [ ] No gate, pump, or operational control changes the stated outlet law.

These four checks document why Qout = Qout(H) is applicable. Until all are checked, routing remains exploratory and constrained screening is locked.

Free-discharge evidence record (required for constrained screening) (?)

Drawing / field inspection / calculation record identifie 5

Screening-limit basis (required for constrained screening)

Source and applicability of Qallow, Hallow, Td, and

These records are preserved in the run report. They document user-provided evidence; the software does not independently verify tailwater, backwater, or project design criteria.

Outlet discharge-coefficient basis

Coefficient mode

Fixed coefficient

Weir Cw preset	Orifice Cd preset
Taiwan practice—sharp-crested rectangular weir (Cw = 1.70)	Standard sharp-edged orifice (Cd = 0.62)

Riser inlet Cr preset	Riser shaft Cd preset
Standard sharp-crested inlet (Cr = 1.70)	Standard sharp-edged shaft (Cd = 0.62)

Fixed values are preliminary screening settings. Select field-calibrated value to enter a project-specific coefficient.

▸ **Coefficient references and applicability**

- [ ] Circular pipe outlet
- [x] Sharp-crested weir 1

Weir coefficient Cw (?)	Weir width b (m) (?)
1.70 2	0.20

Weir crest elevation (m) (?)

0

- [ ] Rectangular opening 3

Opening coefficient Cd (?)	Opening area Ao (m2) (?)
0.62	0.010

Opening height a (m)	Opening width b = Ao/a (m)
0.10	0.100

Opening invert elevation (m) (?)

0.05

BandGate modular workspace

Figure 7. Module 3, outlet hydraulics and free-discharge evidence

1. Enable the sharp-crested weir
2. Weir coefficient Cw, width b, crest elevation
3. Rectangular opening (with the weir it is research inspection)
4. Free-discharge boundary and four checks
5. Two evidence records

8. Module 4: run routing

- Press **"Run routing and evaluation"**. Results appear in Module 6.
- A **grey** button means a field in Modules 1–3 is not valid. The fields to fix are listed above the button. Click one to jump to it (it is outlined in red).
- Common causes: Qp of 0 or less, plan area of 0, weir width or coefficient of 0, an outlet below the floor, or no outlet enabled.



Figure 8. Module 4: hint for a wrong input (Qp = −1)

1. List of fields to fix; click to jump
2. Run button is grey and disabled

9. Module 6: reading the results

- **Check the mass-balance residual ε/Vin first**. It should be close to 0 %. Above 0.5 % the software asks you to resolve the numerical problem first.
- **Peak outflow Qp,o** and **peak ratio RQ**: a smaller RQ means more peak reduction.
- **Event-added detention volume ΔSmax** and **maximum level Hmax**: compare with the allowable level and freeboard.
- **Near-peak flatness FPI95**: a large value means a flat peak, so a single peak time is unreliable. Use the volume-time measures instead.

- Charts: the left chart shows inflow and outflow; the right chart shows water level and storage. "Display duration" sets the time axis.
- The three-step screening and optimization check covers Module 5 results only. After plain routing it shows "screening not yet run".

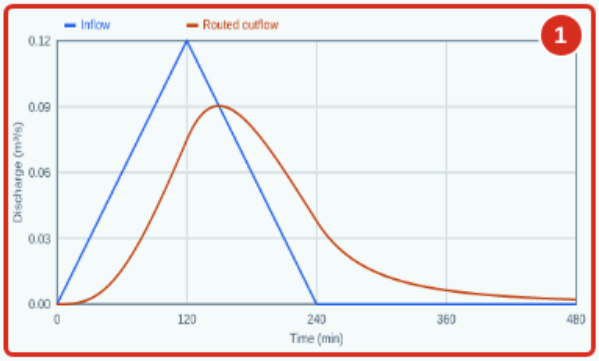
6. Evaluation & decision support

Routing result and engineering interpretation

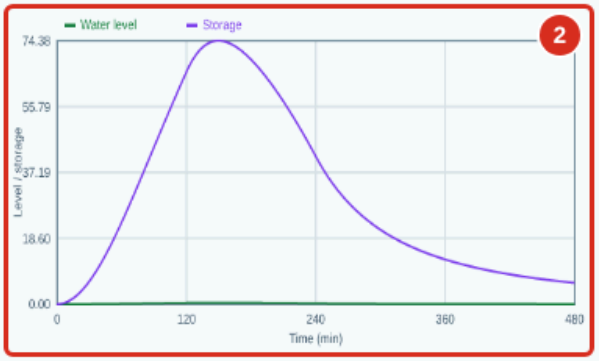
Display duration (min) 120

State routing in S; RK4; Δt = 2 s; 45001 steps

Inflow and routed outflow



Water level and storage



Constrained screening cases

Horizontal: declared candidate plan area. Vertical: maximum detained storage. Green: feasible; gray: constraint failed; red ring: selected feasible screening case. This display is conditional screening, not final design validation.

Output	Value	Engineering meaning
Peak inflow Qp,i	0.1200 m³/s	Declared hydraulic loading 3
Peak outflow Qp,o	0.0903 m³/s	Sum of active outlet components
Peak ratio RQ = Qp,o/Qp,i	0.753	Peak attenuation descriptor
Event-added detention volume ΔSmax	74.385 m³	Maximum storage above the initial storage S0.
Drain-down residual at Te + Td	0.03 %	Declared limit = 1.00% of ΔSmax.
Storage ratio RS = ΔSmax/Vin	0.344	Event-added storage relative to inflow volume
Normalized lag RT	0.124	(tp,o - tp,i) / event duration
Peak water level Hmax	0.413 m	Compare with allowable level and freeboard
Absolute conservation residual	1.259e-10 m³	Absolute difference between cumulative net inflow and storage change
Mass-balance residual ε/Vin	0.000 %	Relative numerical conservation check
Near-peak flatness FPI95	0.006	95% near-peak duration / routed-record duration
Median-volume time	42.6 min	Bulk timing measure
Discharge-centroid time	49.3 min	Bulk timing measure

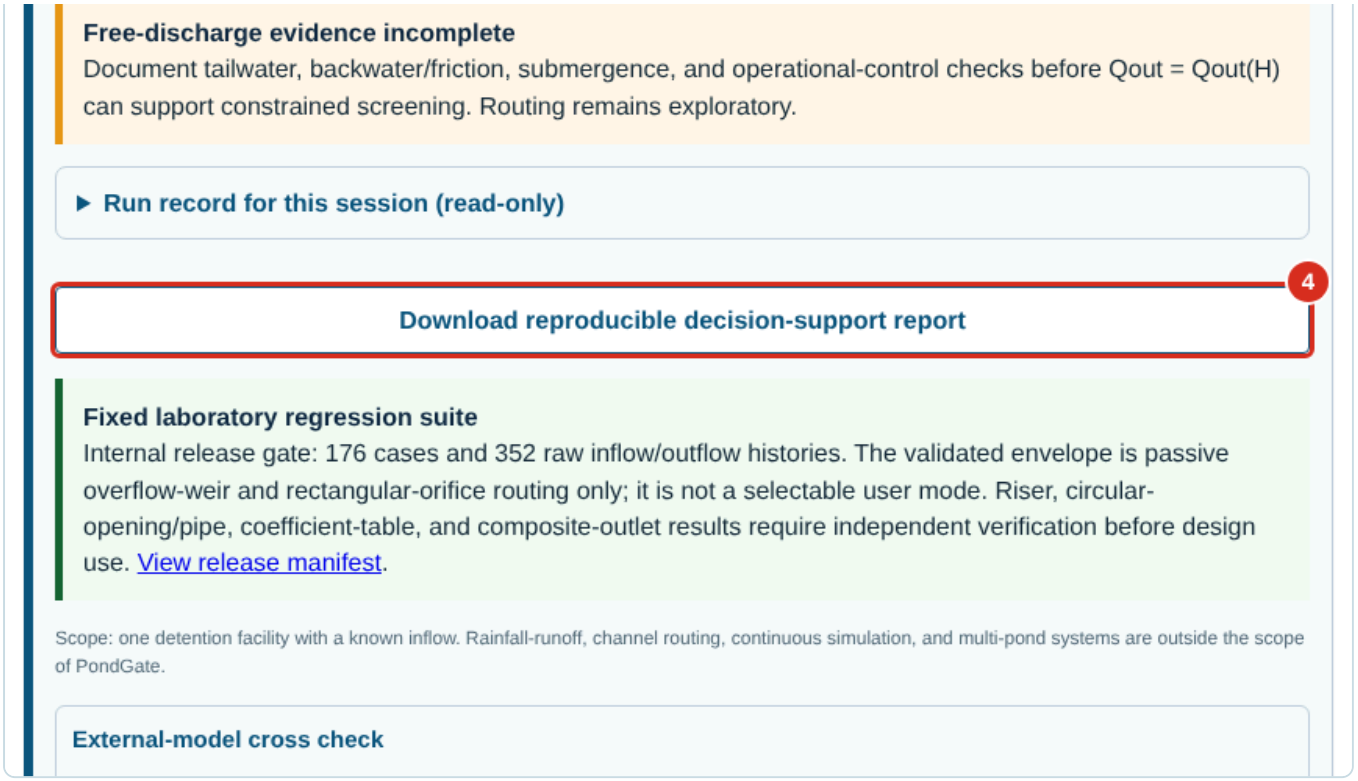


Figure 9. Module 6, routing results

1. Inflow and routed outflow
2. Water level and storage
3. Result table (check the mass-balance residual first)
4. Download the traceable report

10. Module 5: constrained screening and continuous optimization

- Opens only with a **single sharp-crested weir** or a **single rectangular opening**, fixed coefficients, the four free-discharge checks and both records complete.
- Enter the **allowable peak outflow Qallow** (m^3/s) and the **allowable water level Hallow** (m). Both should come from a regulation or design criterion and be cited in the screening-limit record.
- **Maximum drain-down time Td** and **permitted residual percentage**: at Te + Td the event-added storage must not exceed that percentage of ΔSmax.
- The **reference level Href** only defines the dimensionless groups ΠQ and ΠS. It is not automatically a safety limit.
- Search bounds: enter minimum and maximum weir width b or opening area Ao. Each selected size is tested at five equally spaced values.
- After "Run constrained design screening", the page switches to Module 6 to show the result.

- **Continuous optimization** searches plan area A and one outlet size together between the area bounds. The result is not a proven global optimum.
- **No feasible solution**: the software shows "Blocked", the dominant limit (peak outflow, maximum level or drain-down residual) and a "Go to fix" link. Change the pond area or outlet size on engineering grounds. **Do not relax Qallow or Hallow just to obtain a solution.**

5. Constrained design screening

Design screening and constrained routing

RQ = 0.753; RS = ΔSmax/Vin = 0.344; RT = 0.124; ΠH = 0.500; ΠQ(Href) = 2.833; ΠS,Δ(Href) = 0.833; n = 1.5; Href = 1.000 m; Tp/Tr = 1.000. ΠS,Δ = [S(Href) − S0]/Vin is the declared event-added storage control. These are not universal design equations.

Allowable peak outflow (m3/s) (?)

0.080 (1)

Allowable water level (m) (?)

1.00

Outlet configuration for constrained screening

Formal screening is limited to one passive weir or one rectangular opening under confirmed free discharge. Other choices are retained only for research inspection and keep screening locked.

Passive weir only — formal screening (2)

Screening resolution

Standard discrete screen: up to two outlet dimensions

Screening objective

Minimum event-added detention volume ΔSmax

Dimensionless reference water level Href (m) (?)

1.00

Href must use the same datum as H, lie within the declared S(H) domain, and be no lower than H0. It defines ΠQ and ΠS; it is not automatically a safety limit.

Maximum drain-down time Td (h) (?)

24

Permitted residual event storage (%)

1

At Te + Td, event-added storage must not exceed the stated residual percentage of ΔSmax. Every screening case uses the same state equation, constraints and time step.

☑ Screen weir width b

Minimum b (m)

0.05 (3)

Maximum b (m)

1.00

☐ Screen opening area Ao

Minimum Ao (m2)

0.005

Maximum Ao (m2)

0.050

☐ Screen riser diameter D (research inspection only)

Minimum D (m)

0.15

Maximum D (m)

0.60

Standard search: up to two outlet dimensions; other active dimensions remain fixed.
Current Outlet-module dimensions (fixed unless selected for search): b = 0.20 m; Ao = 0.010 m²; D = 0.30 m.

Advanced continuous optimization | global candidate search + local refinement

The formal mode can search two continuous variables: plan area A and the size of one passive outlet (weir width b or opening area Ao). Every candidate reruns the same authoritative RK4 and never writes back to the working inputs. The result is not a proven continuous global optimum and is not final engineering-design validation.

Continuous minimum plan area A (m²)

60

Continuous maximum plan area A (m²)

600

Continuous mode ready: A + b will be searched with the authoritative RK4 router.

▸ **Declared screening bounds**

(4)



Figure 10. Module 5, constrained-screening settings

1. Allowable outflow Qallow and level Hallow
2. Outlet combination for screening
3. Search bounds (minimum and maximum weir width)
4. Run constrained design screening



Selected feasible constrained-screening case — routed

The program evaluated the full visible grid, then performed 2 local refinement passes inside the declared size bounds. The plots above are rerouted with this selected screening case. No screened outlet dimension remains at a declared bound.

Screening item	Result	Interpretation
Outlet configuration	Passive weir only — formal screening	Only listed outlet elements are active.
Plan area A	460 m²	Selected on the declared 20 m² grid.
Weir width b	0.53 m	screened
Event-added detention volume ΔSmax	92.0 m³	Maximum routed storage above the initial storage S0.
Drain-down residual at Te + Td	0.07 / 1.00%	Remaining event-added storage must satisfy the declared drawdown constraint.
Peak outflow / allowable	0.0798 / 0.0800 m³/s	Outlet-peak constraint satisfied.
Peak water level / allowable	0.200 / 1.000 m	Water-level constraint satisfied.
Response indicators	RQ = 0.665; RS = 0.426	Attenuation and event-added storage descriptors.

This is a conditional discrete screening result under the declared bounds. It is not a continuous global optimum or final engineering-design validation.

Figure 11. Screening found a feasible case (shown in Module 6)

1. Selected area, weir width and limit checks

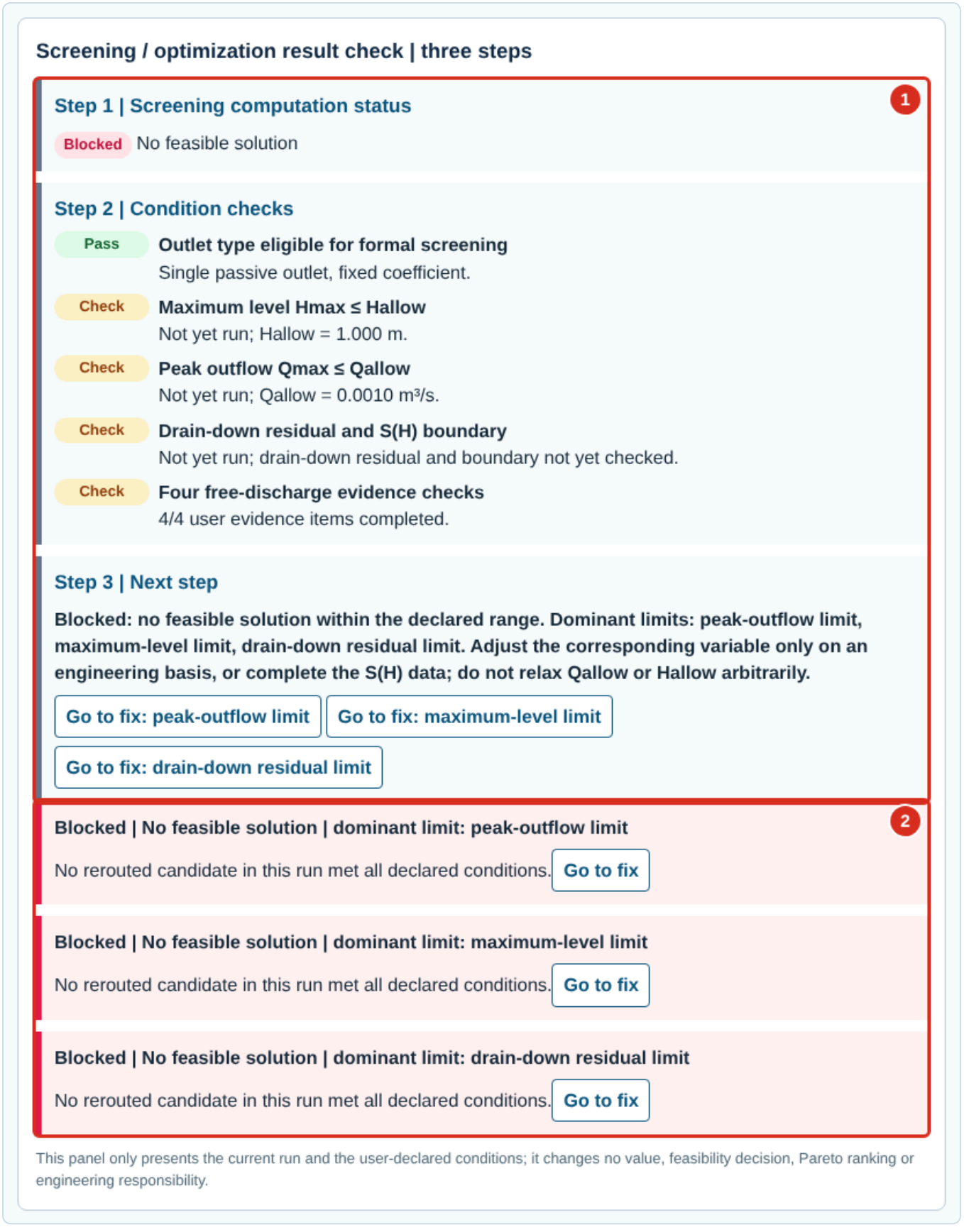


Figure 12. Blocking diagnosis with no feasible solution

1. Three-step check: status, conditions, next step
2. Blocked items with "Go to fix"

11. Reports and run records

- The read-only run record in Module 6 lists the inputs, method, results and limits of this run.
- "Download reproducible decision-support report" saves a text file with inputs, results and scope for your design file.
- When inputs change, the old result becomes invalid. Run again.

12. Comparing with HEC-HMS or SWMM

- In Module 6, "External model cross-check", choose the reference hydrograph (the current routing or public benchmark A-005 to D-005) and the external model.
- Import the external model's **actual simulation result CSV**. The first row holds column names. Required columns are time_s (or time, t) and outflow_m3_s (or qout, qo). A full comparison also needs incremental_storage_m3 (or s, storage).
- The external model must use the same inflow, initial level, S(H), Q(H), coefficients and timing. Agreement means numerical consistency only, not field validation.

External-model cross check

Compare an external result with the current PondGate route or one of four public PondGate benchmarks. The external model must use the same inflow, initial level, S(H), Q(H), coefficients and time settings; this is numerical / implementation cross checking, not field validation.

PondGate reference

Current interactive routing result 1

External model

HEC-HMS 2

Comparison basis

Direct engineering units

External-model reporting interval s

60

Import result CSV

Choose File No file chosen 3

Import only the actual HEC-HMS/SWMM simulation-result CSV. Required for outflow comparison: time_s (or time/t; model_time_min in scaled mode) and outflow_m3_s (or qout/qo; scaled_outflow_m3_s in scaled mode). A complete cross check also requires incremental_storage_m3 (or s/storage). Do not import inflow, curve, model_spec, or PondGate reference-output files. The first row must be headers.

The public package excludes the 176 fixed laboratory regression cases. Selecting A-005, B-005, C-005 or D-005 uses that case's fixed PondGate reference hydrograph directly and does not modify the routing core. Download four cross-check benchmarks

> The current interactive route will be used as reference; run routing first and import the external-model result CSV.

4 **Compare external result**

External-comparison provenance and scope record

Declare the comparison case before importing HEC-HMS/SWMM output. Numerical agreement is not field validation; a public compound-outlet case is also not formal PondGate constrained-screening validation.

Evidence category

PondGate scope-matched: one pond, known inflow, one passive free-discharge outlet

Case identifier

External model / version

Public source URL

Input mapping and scope statement

Load public SWMM detention-pond registry

Download external-output CSV template

The template demonstrates columns and units only; it is not validation output.

Figure 13. External model cross-check

1. PondGate reference hydrograph
2. External model (HEC-HMS / SWMM)
3. Import the external result CSV
4. Compare external results

13. Frequently asked questions

- **The button is grey?** Read the red list above the button and click a field name to fix it.
- **Screening stays locked?** Check that only one outlet is enabled, coefficients are fixed, and the four free-discharge checks and both records are complete.
- **"No feasible solution"?** This is a deliberate block: no design within the current bounds meets all limits. Adjust area or outlet size according to the dominant limit, or widen search bounds that you can justify.
- **Is my data uploaded?** No. Calculation, help and the AI assistant all run in your browser and send nothing.
- **Can a result serve as design approval?** No. It holds only under the declared conditions and needs a professional's judgement.

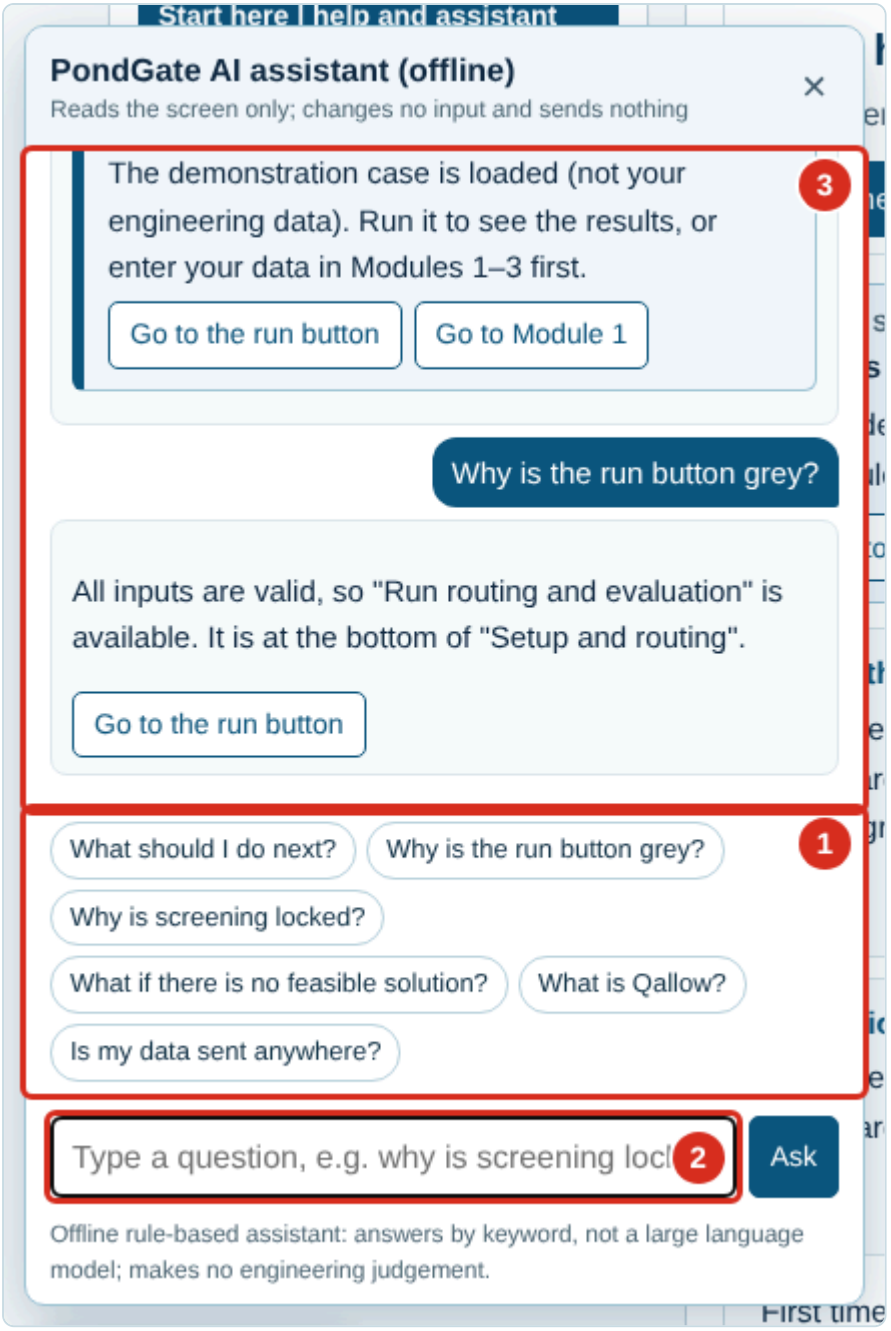


Figure 14. AI assistant (offline)

1. Common-question buttons
2. Type a question
3. Answers and the next-step hint

Glossary

Peak inflow Qp

Largest discharge of the inflow hydrograph (m^3/s). Must be greater than 0.

Time to peak Tp

Time for a triangular inflow to rise from 0 to its peak (min).

Recession time Tr

Time for a triangular inflow to fall from its peak to 0 (min).

Total event duration T

Total length of the inflow hydrograph (min), added up from the segment times. It cannot be typed.

Stage–storage relation S(H)

Relation between water level H and stored volume S. With constant area, S = A × (H − Hfloor).

Plan area A

Pond surface area in constant-area mode (m^2). Must be greater than 0.

Floor elevation Hfloor

Elevation of the pond floor (m). All levels and outlet elevations must share its datum.

Initial level H0

Water level in the pond when the event starts (m). Must not be below Hfloor.

Vertical datum

Common zero for elevations. Floor, levels, crest and invert must share it, or results are wrong.

Weir coefficient Cw

Coefficient in the weir law Q = Cw × b × (H − P)^1.5. The 1.70 preset is for preliminary screening only.

Weir width b

Crest length of the sharp-crested weir (m).

Crest elevation P

Water starts to spill once the level exceeds this elevation (m).

Discharge coefficient Cd

Coefficient in the opening law Q = Cd × Ao × √(2g(H − invert)). Preset 0.62.

Opening area Ao

Area of the rectangular opening (m^2) = height a × width b.

Invert elevation

Lower edge of the rectangular opening (m). Outflow starts above it.

Research inspection

Riser, pipe, coefficient-table or combined-outlet settings. Routing works but constrained screening is locked; independent verification is needed.

Free discharge

Outlet flow depends only on pond level, not on tailwater, backwater, submergence or gate operation. Evidence and the four checks are required before screening.

Two evidence records

"Free-discharge evidence" and "screening-limit source": cite drawing or inspection IDs, checker and date, and the source of limits such as Qallow and Hallow.

RK4

Fourth-order Runge–Kutta method that steps dS/dt = Qin − Qout with a fixed Δt = 2 s.

Mass-balance residual ε/Vin

Difference between cumulative net inflow and storage change, divided by inflow volume. Should be near 0 %; above 0.5 % the result must not be used.

Peak ratio RQ

RQ = peak outflow ÷ peak inflow. Smaller means more peak reduction.

Event-added detention ΔSmax

Largest storage added above the initial storage during the event (m^3).

Storage ratio RS

RS = ΔSmax ÷ inflow volume Vin.

Dimensionless lag RT

RT = (time of peak outflow − time of peak inflow) ÷ event duration.

Maximum level Hmax

Highest pond level during the event (m). Compare with Hallow and freeboard.

Near-peak flatness FPI95

Share of routed time with outflow above 95 % of its peak. A large value means a flat peak, so a single peak time is unreliable.

Allowable peak outflow Qallow

Upper limit of peak outflow in screening (m^3/s), taken from a regulation or design criterion.

Allowable level Hallow

Upper limit of the maximum level in screening (m). Must not be below H0.

Drain-down time Td and residual

At event end Te plus Td, event-added storage must not exceed the stated percentage of ΔSmax.

Reference level Href

Level used only to define the dimensionless groups ΠQ and ΠS (m). Not automatically a safety limit.

Dimensionless groups ΠH, ΠQ, ΠS

Ratios that scale level, outlet capacity and storage by the inflow, used to compare cases. Not general design formulas.

Constrained screening

Reruns routing over the declared bounds to find feasible pond areas and outlet sizes under the declared limits. Not a final design.

Continuous optimization

Continuous search of area A and one outlet size between bounds, then local refinement. Not a proven global optimum.

Blocked / no feasible solution

No candidate within the current limits and bounds meets every condition. The software shows the dominant limit and a correction path, never "pass".

Numerical consistency

PondGate and HEC-HMS or SWMM give close results under identical inputs and physics. Not field calibration or general validation.

176-case regression

Release check that reruns 176 laboratory cases (352 raw series) through the RK4 core. Peak-discharge MARE: A 2.924 %, B 2.954 %, C 4.104 %, D 4.649 %.